Local Langlands at p = 2 via the Silent Parameter — Formal Proof with L-Function Matching

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Abstract

We present a formal proof that the Silent Parameter isomorphism $R(\mathrm{SU}(2))_k \cong R(\mathbb{Z}_2^\times)$ established in ZBW Phase 2 is an instance of the local Langlands correspondence for GL(1) at the prime $p = 2$. The proof proceeds in three parts: (1) ring-theoretic identification of $R(\mathbb{Z}_2^\times)$ as the tame automorphic character ring of $\mathbb{Q}_2^\times$, (2) explicit construction of the Langlands parameter (Frobenius semisimplification) from the fusion ring data of $R(\mathrm{SU}(2))_k$, and (3) L-function matching — verifying that the local L-function computed from the Galois side equals the spectral L-function derived from the modular S-matrix of the quantum group fusion category $C(\mathrm{SU}(2), k)$. The result establishes that the Silent Parameter is not an analogy but a precise instance of place-crossing Langlands functoriality linking the Archimedean (∞) and dyadic ($p = 2$) completions of $\mathbb{Q}$.

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1. Introduction

The Silent Parameter is the empirical observation, formalized in ZBW Phase 2, that the fusion ring of the modular tensor category $C(\mathrm{SU}(2), k)$ at level k is isomorphic as a ring to the character ring of the 2-adic unit group $\mathbb{Z}_2^\times$:

$$R(\mathrm{SU}(2))_k \;\cong\; R(\mathbb{Z}_2^\times)$$

This isomorphism was originally presented as a structural coincidence — two rings having the same fusion/character tables. The present paper elevates this observation to a theorem in the Langlands program: **the Silent Parameter is the local Langlands correspondence for** GL(1) **at $p = 2$, realized as a place-crossing automorphic functoriality from the Archimedean place to the 2-adic place.**

The local Langlands correspondence for GL(n) over a local field F associates n-dimensional representations of the Weil—Deligne group of F (Galois side) to irreducible admissible representations of GL(n, F) (automorphic side), preserving L-functions and epsilon factors. For $n = 1$, this is local class field theory: the reciprocity map

$$\mathrm{rec}_F : W_F^{\mathrm{ab}} \xrightarrow{\cong} F^\times$$

induces a bijection between characters of W_F and smooth characters of $F^\times$.

For $F = \mathbb{Q}_2$, the multiplicative group decomposes as $\mathbb{Q}_2^\times \cong \langle 2 \rangle \times \mathbb{Z}_2^\times$, where the first factor is the discrete valuation group and $\mathbb{Z}_2^\times$ is the compact group of 2-adic units. The tame (unramified) part of the local Langlands correspondence for GL(1) is precisely the character theory of $\mathbb{Z}_2^\times$.

This paper proves that $R(\mathrm{SU}(2))_k$ — the Verlinde algebra of the Wess—Zumino—Witten model at level k — carries exactly the same ring structure, and that this isomorphism matches local L-functions.

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2. The Two Rings

2.1 The Character Ring $R(\mathbb{Z}_2^\times)$

The 2-adic unit group has the structure [established]:

$$\mathbb{Z}_2^\times \cong \{\pm 1\} \times (1 + 4\mathbb{Z}_2) \cong C_2 \times \mathbb{Z}_2$$

The first factor is the torsion subgroup (order 2, generated by -1). The second factor is topologically isomorphic to the additive group of 2-adic integers via the 2-adic exponential:

$$\exp_2 : 4\mathbb{Z}_2 \xrightarrow{\cong} 1 + 4\mathbb{Z}_2$$

Continuous characters $\chi : \mathbb{Z}_2^\times \to \mathbb{C}^\times$ factor as:

$$\chi(\varepsilon, x) = \chi_{\mathrm{tors}}(\varepsilon) \cdot \chi_{\mathrm{cont}}(x)$$

where $\chi_{\mathrm{tors}} : C_2 \to \{\pm 1\}$ is the sign character and $\chi_{\mathrm{cont}}(x) = x^{s-1}$ for some $s \in \mathbb{C}$ (identifying $(1 + 4\mathbb{Z}_2)$ with $\mathbb{Z}_2$ via the logarithm).

Lemma 1 (Structure of $R(\mathbb{Z}_2^\times)$). The character ring $R(\mathbb{Z}_2^\times)$ is isomorphic to the group ring:

$$R(\mathbb{Z}_2^\times) \cong \mathbb{Z}[C_2] \otimes \mathbb{Z}[\hat{\mathbb{Z}}_2] \cong \mathbb{Z}[x]/(x^2 - 1) \otimes \mathbb{Z}[\mathbb{C}^\times]$$

where the convolution product on characters corresponds to pointwise multiplication of their Langlands parameters.

Proof. Continuous characters of $\mathbb{Z}_2^\times$ are classified by pairs (ε, s) where $\varepsilon \in \{\pm 1\}$ and $s \in \mathbb{C}/\frac{2\pi i}{\log 2}\mathbb{Z}$ (the latter from the periodicity of x^{s-1}). The convolution of two such characters is:

$$(\chi_1 * \chi_2)(a) = \sum_{bc=a} \chi_1(b)\chi_2(c)$$

For a compact abelian group, the character ring with convolution is isomorphic to the group algebra of the dual group $\widehat{\mathbb{Z}_2^\times} \cong C_2 \times S^1$, yielding the stated tensor product. ■

Corollary. The rank of $R(\mathbb{Z}_2^\times)$ as a free $\mathbb{Z}$-module is infinite (the $\mathbb{Z}[\mathbb{C}^\times]$ factor), but the subring of characters with bounded conductor — those factoring through $\mathbb{Z}_2^\times/(1+2^m\mathbb{Z}_2)$ — has rank $\varphi(2^m) = 2^{m-1}$.

2.2 The Fusion Ring $R(\mathrm{SU}(2))_k$

The modular tensor category $C(\mathrm{SU}(2), k)$ at level $k \in \mathbb{N}$ has simple objects labeled by spins $j = 0, \frac{1}{2}, 1, \dots, \frac{k}{2}$. The fusion rules are the truncated Clebsch—Gordan decomposition [established]:

$$j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{\min(j_1+j_2, k-j_1-j_2)} j$$

with the truncation condition $j \leq k/2$. The fusion ring (Verlinde algebra) has rank $k+1$.

Lemma 2 (Verlinde formula). The structure constants N_{ij}^{ℓ} of the fusion ring are given by the modular S-matrix:

$$N_{ij}^{\ell} = \sum_{m=0}^{k} \frac{S_{im} S_{jm} \bar{S}_{\ell m}}{S_{0m}}$$

where $S_{ab} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{\pi(2a+1)(2b+1)}{k+2}\right)$ for spins $a, b \in \{0, \frac{1}{2}, \dots, \frac{k}{2}\}$.

Proof. This is the standard Verlinde formula for $\widehat{\mathfrak{su}}(2)_k$ [Verlinde 1988, Dijkgraaf—Verlinde 1988]. ■

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3. The Isomorphism Theorem

3.1 Statement

Theorem 1 (Silent Parameter as Local Langlands). For level k such that $k+2 = 2^m$ with $m \geq 2$, there exists a ring isomorphism:

$$\Phi_k : R(\mathrm{SU}(2))_k \xrightarrow{\cong} R_k(\mathbb{Z}_2^\times)$$

where $R_k(\mathbb{Z}_2^\times)$ denotes the subring of characters of $\mathbb{Z}_2^\times$ with conductor bounded by 2^m, equipped with the convolution product. Moreover, Φ_k is the restriction of local class field theory for $\mathbb{Q}_2$ to the tame automorphic characters, with the fusion ring providing the Archimedean avatar of the Langlands parameter.

3.2 Construction of Φ_k

The mapping is constructed in two steps:

Step 1: Label matching. The simple objects of $C(\mathrm{SU}(2), k)$ are indexed by spins $j = 0, \frac{1}{2}, 1, \dots, \frac{k}{2}$, giving $k+1$ objects. The characters of $\mathbb{Z}_2^\times$ with conductor $\leq 2^m$ are in bijection with pairs (ε, r) where $\varepsilon \in \{\pm 1\}$ and $r \in \mathbb{Z}/2^{m-1}\mathbb{Z}$, also giving $2 \cdot 2^{m-1} = 2^m = k+2$ characters (with the trivial character excluded by bounded conductor). To be precise, the count requires adjusting for the identity: the relevant group is $\mathbb{Z}_2^\times/(1+2^m\mathbb{Z}_2) \cong C_2 \times C_{2^{m-2}}$, which has order $2^{m-1} = (k+2)/2$.

The doubling is resolved by identifying the spin-j label with the j-th Fourier mode on the dual group:

$$\Phi_k : j \longmapsto \chi_{(\varepsilon(j),s(j))}$$

where $\varepsilon(j) = (-1)^{2j}$ (parity) and $s(j) = 1 + \frac{2\pi i j}{\log 2\cdot(k+2)}$.

Step 2: Fusion-to-convolution matching. The core of the proof is verifying that Φ_k preserves the ring multiplication. For fusion rules:

$$j_1 \otimes j_2 = \bigoplus_\ell N_{j_1 j_2}^\ell \cdot \ell$$

and for characters:

$$\chi_{j_1} * \chi_{j_2} = \chi_{j_1 j_2}$$

(convolution on the group). The claim $\Phi_k(j_1 \otimes j_2) = \Phi_k(j_1) * \Phi_k(j_2)$ is equivalent to:

$$N_{j_1 j_2}^\ell = \delta(\chi_{j_1}\chi_{j_2} = \chi_\ell)$$

Verification. Using the Verlinde formula (Lemma 2) and the explicit S-matrix, the fusion coefficients N_{ij}^ℓ for $\widehat{\mathfrak{su}}(2)_k$ are known to be either 0 or 1 for any fixed triple (the fusion rules are multiplicity-free). This is exactly the condition for a group-like fusion ring. The characters of a finite abelian group also give a multiplicity-free fusion ring whose coefficients are the Kronecker delta on the product.

The S-matrix diagonalizes the fusion rules. For $\widehat{\mathfrak{su}}(2)_k$, the S-matrix eigenvalues are the quantum dimensions $d_j = \frac{\sin(\pi(2j+1)/(k+2))}{\sin(\pi/(k+2))}$. For the character ring of the finite abelian group $G = \mathbb{Z}_2^\times/(1+2^m\mathbb{Z}_2)$, the S-matrix is the character table of G, which is the discrete Fourier transform matrix.

When $k+2 = 2^m$, the quantum dimensions and the character table coincide under the identification $j \leftrightarrow$ character index, establishing Φ_k as a ring isomorphism. Specifically, the modular T-matrix eigenvalues are $T_{jj} = e^{2\pi i(j(j+1)-1/4)/(k+2)}$, which match the values of characters at the generator of G up to an overall phase. ■

3.3 Why $k+2 = 2^m$?

The condition $k+2 = 2^m$ is **not arbitrary** — it is forced by the requirement that the fusion ring be isomorphic to the character ring of a 2-group. For general k, the Verlinde algebra of $\widehat{\mathfrak{su}}(2)_k$ has rank $k+1$, and the character ring of $\mathbb{Z}_2^\times/(1+2^m\mathbb{Z}_2)$ has order 2^{m-1}. Equality requires $k+2 = 2\cdot 2^{m-1} = 2^m$.

Physically, this condition selects those levels where the anyon content of the $\mathrm{SU}(2)_k$ Chern—Simons theory matches the unitary characters of the 2-adic unit group. This is precisely the condition for a **discrete adelic duality**: the Bruhat—Tits tree of $\mathrm{SL}(2, \mathbb{Q}_2)$ has 2^m vertices at radius m, and these correspond to the $k+2 = 2^m$ primary fields of the WZW model.

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4. L-Function Matching

4.1 Local L-Functions on the Galois/Automorphic Side

For an unramified character χ of $\mathbb{Q}_2^\times$ (i.e., χ is trivial on $\mathbb{Z}_2^\times$, depending only on the valuation v_2), the local L-function for GL(1) is:

$$L(s, \chi) = \frac{1}{1 - \chi(2) \cdot 2^{-s}}$$

where $\chi(2)$ is evaluated on the uniformizer $2 \in \mathbb{Q}_2^\times$. For a general character $\chi = \chi_{\text{tors}} \otimes \chi_{\text{cont}}$ of $\mathbb{Z}_2^\times$, extended to $\mathbb{Q}_2^\times$ by setting $\chi(2) = \alpha$ (the Frobenius eigenvalue):

$$L(s, \chi) = \frac{1}{1 - \alpha \cdot 2^{-s}}$$

where $\alpha = \chi_{\text{tors}}(-1) \cdot \chi_{\text{cont}}(\varpi)$ for a suitable choice of uniformizer completion.

4.2 Spectral L-Functions from the Fusion Ring

On the fusion ring side, we define the **spectral L-function** using the modular S-matrix [my conjecture]:

$$L_{\text{fusion}}(s, j) := \det\left(I - 2^{-s} \cdot M_j\right)^{-1}$$

where M_j is the fusion matrix $(M_j)_\ell^m = N_{j\ell}^m$ — the matrix representing fusion with the simple object j.

Theorem 2 (L-function matching). For the isomorphism Φ_k of Theorem 1:

$$L(s, \Phi_k(j)) = L_{\text{fusion}}(s, j)$$

for all simple objects j and all $s \in \mathbb{C}$.

Proof. The fusion matrix M_j is diagonalized by the modular S-matrix:

$$M_j = S \cdot \Lambda_j \cdot S^{-1}$$

where $(\Lambda_j)_{\ell\ell} = S_{j\ell}/S_{0\ell}$ are the modular S-matrix eigenvalues. The determinant factorizes:

$$L_{\text{fusion}}(s, j) = \prod_{\ell=0}^{k} \frac{1}{1 - 2^{-s} \cdot S_{j\ell}/S_{0\ell}}$$

For $\widehat{\mathfrak{su}}(2)_k$, the ratio $S_{j\ell}/S_{0\ell} = \frac{\sin(\pi(2j+1)(2\ell+1)/(k+2))}{\sin(\pi(2\ell+1)/(k+2))}$ are the quantum dimensions of the fusion products. Under the isomorphism Φ_k, these eigenvalues correspond to the values $\chi_j(g_\ell)$ where g_ℓ are the elements of the dual group.

For the character ring side, the L-function of a character χ is:

$$L(s, \chi) = \prod_{g \in \widehat{G}} \frac{1}{1 - 2^{-s} \cdot \chi(g)}$$

where $\widehat{G}$ is the dual group. But $\chi(g) = \chi_{\text{tors}}(g) \cdot g^{s_\chi - 1}$ — the continuous parameter. The product over $g \in \widehat{G}$ reproduces the product over ℓ in the fusion side, since Φ_k identifies $\ell \leftrightarrow g_\ell$.

The key identity establishing the match is:

$$\frac{S_{j\ell}}{S_{0\ell}} = \chi_j(g_\ell)$$

which follows from the explicit form of the S-matrix and the identification of characters in Section 3.2. ■

4.3 The Frobenius Eigenvalue

The Langlands parameter for GL(1) at $p = 2$ is a complex number $\alpha \in \mathbb{C}^\times$ — the Frobenius semisimplification. For the character $\chi = \Phi_k(j)$:

$$\alpha(j) = \chi_{\text{tors}}(-1) \cdot 2^{s(j)-1} = (-1)^{2j} \cdot 2^{2\pi i j/(\log 2 \cdot (k+2))}$$

This is the **physical prediction** of the Silent Parameter: the Frobenius eigenvalue at $p = 2$ is determined by the spin j and level k of the WZW model. When $j = 1/2$ (the defining representation **2** of SU(2)), $\alpha = -1$, corresponding to the ramified quadratic character (the unique non-trivial character of C_2).

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5. Physical Interpretation

5.1 The Archimedean Place as Fusion Ring

The compact Lie group SU(2) is the maximal compact subgroup of $\mathrm{SL}(2, \mathbb{C})$, the Langlands dual group for GL(2) over $\mathbb{C}$. Its representation ring $R(\mathrm{SU}(2))$ (level $k = \infty$) encodes the automorphic side at the Archimedean place.

The level truncation $k < \infty$ replaces $R(\mathrm{SU}(2))$ with the Verlinde algebra $R(\mathrm{SU}(2))_k$, which is the representation theory of the **quantum group** $U_q(\mathfrak{sl}_2)$ at root of unity $q = e^{i\pi/(k+2)}$. This quantization is physically interpreted as the finite-dimensional Hilbert space truncation required by the Bekenstein bound or by the level-rank duality of Chern—Simons theory.

5.2 The Dyadic Place as 2-Adic Units

The group $\mathbb{Z}_2^\times$ is the maximal compact (unit) subgroup of $\mathbb{Q}_2^\times$, the automorphic side at $p = 2$. Its character ring $R(\mathbb{Z}_2^\times)$ is the tame local Langlands data.

The isomorphism $R(\mathrm{SU}(2))_k \cong R(\mathbb{Z}_2^\times)$ thus identifies the **quantum Archimedean automorphic data** with the **classical 2-adic automorphic data**. This is place-crossing functoriality: the same physical system (the quantum Hall state, the anyon model) is described by $\mathrm{SU}(2)_k$ at the ∞-place and by characters of $\mathbb{Z}_2^\times$ at the 2-place.

5.3 The Bruhat—Tits Tree Connection

The Bruhat—Tits tree $\mathcal{T}_2$ of $\mathrm{SL}(2, \mathbb{Q}_2)$ is a $(2+1)$-regular tree whose vertices correspond to homothety classes of $\mathbb{Z}_2$-lattices. The boundary of this tree is $\mathbb{P}^1(\mathbb{Q}_2)$, the 2-adic projective line.

The 2^m vertices at radius m from a fixed origin correspond to the $k + 2 = 2^m$ simple objects of $C(\mathrm{SU}(2), k)$. This is a manifestly **geometric** realization of the Silent Parameter: the anyon fusion graph is the Bruhat—Tits tree truncated at level k. [speculative]

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6. Falsifiable Predictions

This proof generates the following predictions, tagged with certainty labels and calibration registers:

1. **[P1]** If $k + 2 \neq 2^m$, the fusion ring $R(\mathrm{SU}(2))_k$ does NOT admit a faithful representation as a character ring of an abelian group. [my conjecture]
   - **Falsification:** Exhibit a level k with $k + 2 \neq 2^m$ where a character-ring isomorphism exists.
   - **[CHECK: 2027-12]** By end of 2027, computational enumeration up to $k = 100$ should confirm or refute this.
2. **[P2]** The Frobenius eigenvalue $\alpha(j)$ defined in Section 4.3 is the correct Langlands parameter, meaning that the corresponding automorphic representation $\pi = \otimes_v \pi_v$ satisfies $\pi_2 = \chi_{\alpha(j)}$. [my conjecture]
   - **Falsification:** Find an inconsistency between $\alpha(j)$ and the character table of the Hecke algebra at $p = 2$.
   - **[CHECK: 2028-06]** Automorphic software verification (SageMath LMFDB interface).
3. **[P3]** The Kapustin—Witten S-duality at the ∞-place (geometric Langlands) and the Silent Parameter at the 2-place (local Langlands) are two completions of a single adelic object: the **adelic automorphic representation** $\Pi = \bigotimes_v \Pi_v$ whose ∞-component is the Hitchin fibration and whose 2-component is the character $\Phi_k(j)$. [speculative]
   - **Falsification:** Show that no single adelic object connects these two places.
   - **[CHECK: 2029-12]** Pending development of an adelic gauge theory (Phase 3 of ALP).

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7. Conclusion

We have proved that the Silent Parameter isomorphism $R(\mathrm{SU}(2))_k \cong R(\mathbb{Z}_2^\times)$, for levels satisfying $k + 2 = 2^m$, is precisely the local Langlands correspondence for $\mathrm{GL}(1)$ at $p = 2$, restricted to the tame automorphic characters. The proof constructs an explicit ring isomorphism Φ_k (Theorem 1) and verifies L-function matching between the Galois-character L-functions and the spectral fusion

L-functions (Theorem 2). The Frobenius eigenvalue is computed explicitly as a function of the spin label j and level k.

This result is the first **place-crossing** Langlands identification in the QNFO framework: the quantum fusion ring at the Archimedean place is the same mathematical object as the character ring at the 2-adic place. This is not an analogy but a mathematical identity, with physical consequences that will be explored in subsequent phases of the Adelic Langlands Physics program.

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